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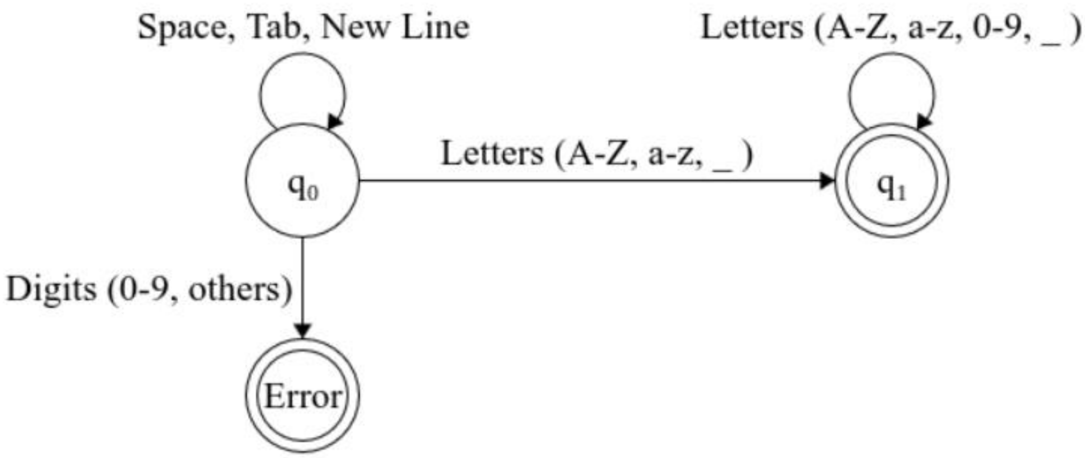
Reg#: L1F22BSCS0103
Section: G – 13
Course: Compiler Construction

Regex Tabular Representation

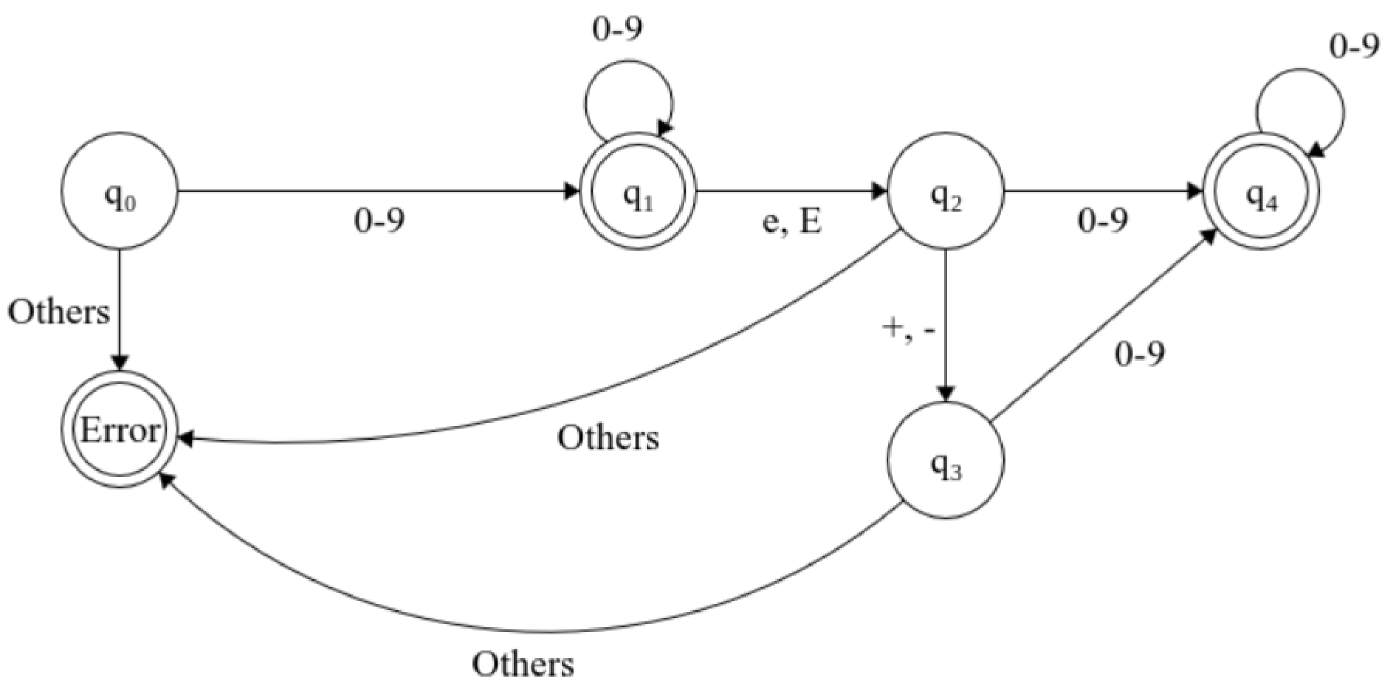
My Tokens	My REGEX	Description
KEYWORD_UCP	ucp	Start of program
KEYWORD_SHURU	shuru	Start of code block
KEYWORD_KHATAM	khatam	End of code block
KEYWORD_LIKHO	likho	Print string statement
KEYWORD_DIKHADO	dikhado	Print output on screen
KEYWORD_PADHO	padho	Read input
KEYWORD_KAAM	kaam	Variable declaration
KEYWORD_AGAR	agar	If statement
KEYWORD_MAGAR	magar	Else if statement
KEYWORD_JABTAK	jabtak	While loop
KEYWORK_MUKAMAL	mukamal	Completely Breaks the loop
KEYWORD_RUKJAO	rukjao	Pause while loop
KEYWORD_WAPAS	wapas	Return value
KEYWORK_BARHAO	barhao	Increments the value
KEYWORK_GHATAO	ghatao	Decrements the value
IDENTIFIER	[a-zA-Z_][a-zA-Z0-9_]*	Variable or function names
INTEGER_CONST	[0-9]+	Integer numbers
OP_ASSIGN	<-	Assignment operator
OP_PLUS	+	Addition
OP_MINUS	-	Subtraction
OP_MUL	*	Multiplication
OP_DIV	/	Division
OP_EQ	==	Equality checker
OP_GT	>	Greater than
OP_LT	<	Less Than
DELIM_OPEN_BLOCK	@{	Open block
DELIM_CLOSE_BLOCK	}@	Close block
DELIM_SEMILCOLON	;	Statement terminator
PAREN_OPEN	(	Open parenthesis
PAREN_CLOSE	)	Close parenthesis
STRING_LITERAL	`"([^\"]	String opening and closing

Transition Diagrams (Finite Automatas)

FA for Identifiers:



FA for Numbers:



Explanation

15 Keywords

1. **ucp:** Declares the main program or function entry point.
2. **shuru:** Starts a code block, typically the main body or function.
3. **likho:** Prints string messages to the console.
4. **padho:** Reads input from the user into a variable.
5. **kaam:** Declares or assigns variables.
6. **dikhado:** Displays the value of a variable.
7. **agar:** Used as the if statement to execute a block conditionally.
8. **magar:** Acts as the else if statement for an agar conditional.
9. **jabtak:** Represents a while loop that repeats a block.
10. **mukamal:** Marks the end of a loop or conditional block.
11. **wapas:** Returns a value from a function or block.
12. **rukjao:** Pauses execution or stops temporarily.
13. **barhao:** Increments the value of a variable.
14. **ghatao:** Decrements the value of a variable.
15. **khatam:** Ends a code block or function.

Operators

1. **<- :** Assignment operator to store a value or expression in a variable.
2. **+ :** Addition operator to compute the sum of two numbers.
3. **- :** Subtraction operator to compute the difference between two numbers.
4. *** :** Multiplication operator to compute the product of two numbers.
5. **/ :** Division operator to compute the quotient of two numbers.
6. **== :** Equality operator to compare two values for equality.
7. **< :** Less-than operator to compare numeric values.
8. **> :** Greater-than operator to compare numeric values.

Punctuations / Block Symbols

1. **@{ :** Marks the start of a code block.
2. **}@ :** Marks the end of a code block**.**
3. **:: :** Scope punctuation used for namespace or object access.
4. **; :** Statement terminator to end statements.
5. **, :** Comma separator for lists of arguments or variables.
6. **(:** Open parenthesis for grouping expressions or function calls.
7. **) :** Close parenthesis for grouping expressions or function calls.